LESSON PLAN

MOCKUP SQUADRON ENVIRONMENT (MSE) (24)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	21	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Installation and operation of contactor
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Installation and operation of contactor.
5. Specific Objectives	24.1.1.1 Trainee will be able to explain the operation of Polarized relay and contacts.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of contactor installation and operation; Safety: De-energize circuit before handling; Supervision: Instructor monitors wiring and testing procedures.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a contactor in an electrical circuit?
(b) How does a polarized relay differ from a standard relay?
(c) Why is it critical to de-energize a circuit before installing a contactor?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Construction of polarized relay • Operation of polarized relay contacts • Installation steps for contactor • Safety precautions during installation • Testing contactor operation	L & Ds	PPT, IB	15 min
(b)	• Wiring configuration for contactor • Coil energization sequence • Contact closure and arc suppression • Troubleshooting common faults • Maintenance and inspection procedures	L & Ds	PPT, PDF	20 min
(c)	• Verification of contactor operation • Documentation of installation • Review of safety protocols • Q&A session on operation principles • Preparation for next lesson topic	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of the polarized relay in controlling the contactor coil.
(b) Describe the steps taken to ensure safe installation of the contactor.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between a Wheatstone bridge and a ratio meter coil, as covered in the next lesson plan (LP 03), focusing on their application in temperature measurement.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Measurement of temperature using Wheatstone bridge, ratio meter coil and thermocouple.	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min